

BEFORE

Infectious diseases have a long history, and the first accounts date back to ancient times.

SMALLPOX IN EGYPT

The first known records of infectious diseases include the mummified remains of Egyptian pharaoh **Rameses V** dating back to c. 1140 BCE. His head appears to show skin lesions similar to those caused by **smallpox**.

EARLY VACCINATION

In ancient China it has been claimed that a lucky day was chosen on which to blow crusts from the skin of a **smallpox** sufferer into a patient's nose through a tube or quill. If the Chinese did inhale a powdered material from the sores of smallpox victims, this would be the **first recorded form of vaccination**.

PESTILENCE IN GREECE

The ancient Greek historian **Thucydides** (c. 460–401 BCE) recorded in painstaking detail the symptoms of the **great pestilence of 430–429 BCE**, which devastated Athens, “so that it may be recognized by medical men if it recurs.”

Despite catching the disease himself, Thucydides survived to leave a **harrowing account of the death and despair** that the disease caused, accounting as it did for some 60,000 lost lives—a quarter of the population of Athens at that time.

FIRST EUROPEAN BUBONIC PLAGUE

From c. 542 CE a bubonic plague had reached Egypt and the Roman Empire, from Arabia: it became known as the **Plague of Justinian** (after the emperor Justinian I). Thus the Black Death was not the first such plague in Europe.



THUCYDIDES

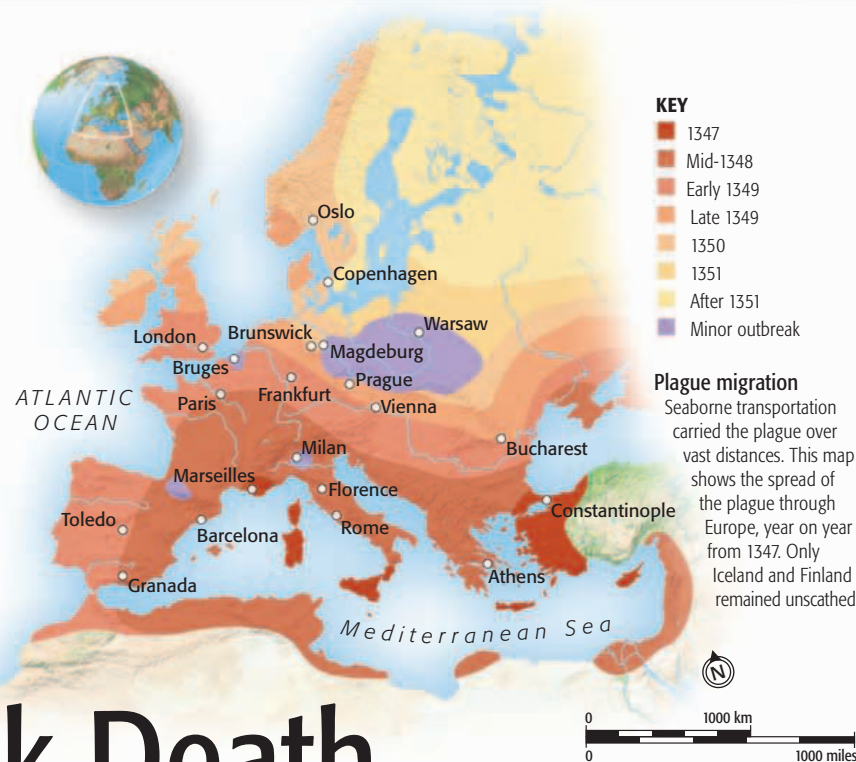
60 PERCENT of Europe's population died from the plague in the 14th century, according to some of the higher estimates, with densely populated urban areas being the worst affected.

HOW WE KNOW

YERSINIA PESTIS BACTERIUM

Medical researchers have extracted genetic evidence of the DNA code of the sausage-shaped *Yersinia pestis* bacterium (see right) from several plague burial sites in French and other cemeteries from the period 1348–1590. Historians studying medieval parish registers have also uncovered abnormally high mortality rates, which along with maritime records and contemporary accounts detailing symptoms, locations, and dates, have brought together random pieces of the plague's timetable and destinations.

The bubonic plague that came to be known as the Black Death spread to Europe from central Asia, where it had already left a trail of devastation. It may have first reached Europe in the fall of 1346, when an army of Muslim Tartars laid siege to the flourishing Christian enclave of Caffa (the modern-day Ukrainian port of Feodosiya) on the Silk Road (see pp.184–85) and plague struck the besieging troops. Stories of infected corpses being catapulted over the walls into Caffa to spread the plague are probably unreliable. What is not disputed is that infected fleas on rats accompanied the besieged Christian merchants when they fled by boat back to the Mediterranean via Constantinople in 1347, and so they



The Black Death

Towns and villages of medieval Europe became littered with corpses, and death carts carried so many bodies that they could only be buried in deep pits. The medieval world had very little protection against a plague that was to wipe out one in three, and maybe more, of the population.

took the bubonic plague with them. Thus the most virulent epidemic then known, and still ever recorded, spread rapidly—and fatally—through much of the Western world.

Diverse strains

Plague is caused by a bacterium called *Yersinia pestis* (see below), which is carried by fleas on wild rodents. Transfer of bacterial infection occurs when the fleas feed on human blood.

There were three variants of the plague: bubonic, in which the patient develops buboes, or swellings, of the neck, groin, or armpit glands and from which this variant got its name; pneumonic, or blood-coughing, in which the lungs are infected because the infection is carried in the air and

inhaled; and septicemic, or blood-poisoning, in which the bacteria attack the blood system itself.

The bubonic variant was the most dramatic and obvious form of Black Death, as described by contemporaries such as the Italian writer Giovanni Boccaccio, who wrote of “certain swellings in the groin or the armpits ... some as large as a common apple.” It took 3–5 days to incubate, and then within a further three days caused death in 80 percent of victims.

The pneumonic strain, with or without the boils, was 90–95 percent lethal within just a few days (today, with treatment, that percentage would probably survive) and could be transmitted directly from person to person—by sneezing, for example. The septicemic variant, though the least common, led to virtually instantaneous death. This most virulent form also caused a victim's skin to turn dark purple, almost black, and may account for the plague's epithet “black”—though this term was not used at the time.

Trail of death

How many perished under its macabre shadow is not known, but it is estimated that in Europe more than 25 million died from this plague pandemic. Other estimates rise to 50 million out of a population of 80 million in the 1300s. The plague returned repeatedly over subsequent generations, but never again with such devastating loss of life.

AFTER

The Black Death had a huge impact on medieval society. But it was not the last plague to strike Europe, and there may be others in the future.

AFTEREFFECTS

The traumatic decline in population led to a shortage of labor and a sudden rise in prices, as well as **irreversible social changes**. The European population did not recover until the 16th century. Over the next four centuries, another **nine major plague pandemics** hit Europe.

MODERN PLAGUES

Precautionary measures now help to control the spread of disease. In Hong Kong in 2003, for example, tissue-paper masks were used to avoid contact with **SARS (Severe Acute Respiratory Syndrome)**—a variety of pneumonic plague. Scientists predict that drug-resistant infections may cause devastating pandemics in the future.



SARS: MODERN PLAGUE

Dance of death

The Black Death touched every level of medieval society, lords and ladies, clergy, and the poor alike. This *Danse Macabre*—a reminder of the universality of death—was a recurrent moral and artistic theme in post-plague Europe.

